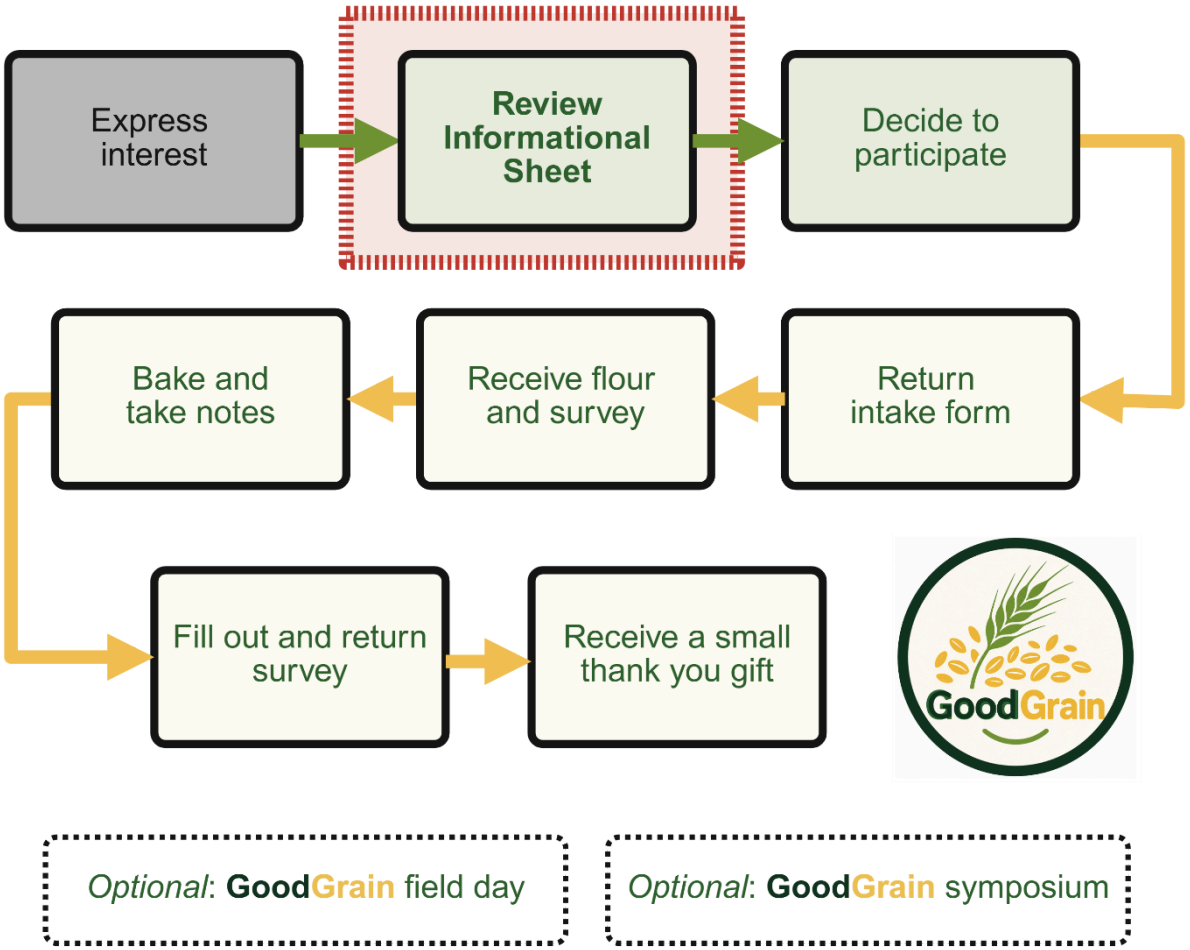
GoodGrain Home Baking Evaluation

agro-nichols-goodgrain.share.connect.posit.cloud/index-en.html

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Informational Sheet

This sheet is meant to give you more information about the study, so you can decide if you would like to participate.



Reading this *Informational Sheet* is meant to aid you in your decision of whether or not to participate in the GoodGrain Home Baking Evaluation. Attendance to the GoodGrain field day GoodGrain and symposium are optional, and are not required for participation in the Home Baking Evaluation. Image made by Virginia Anne Nichols using Biorender.

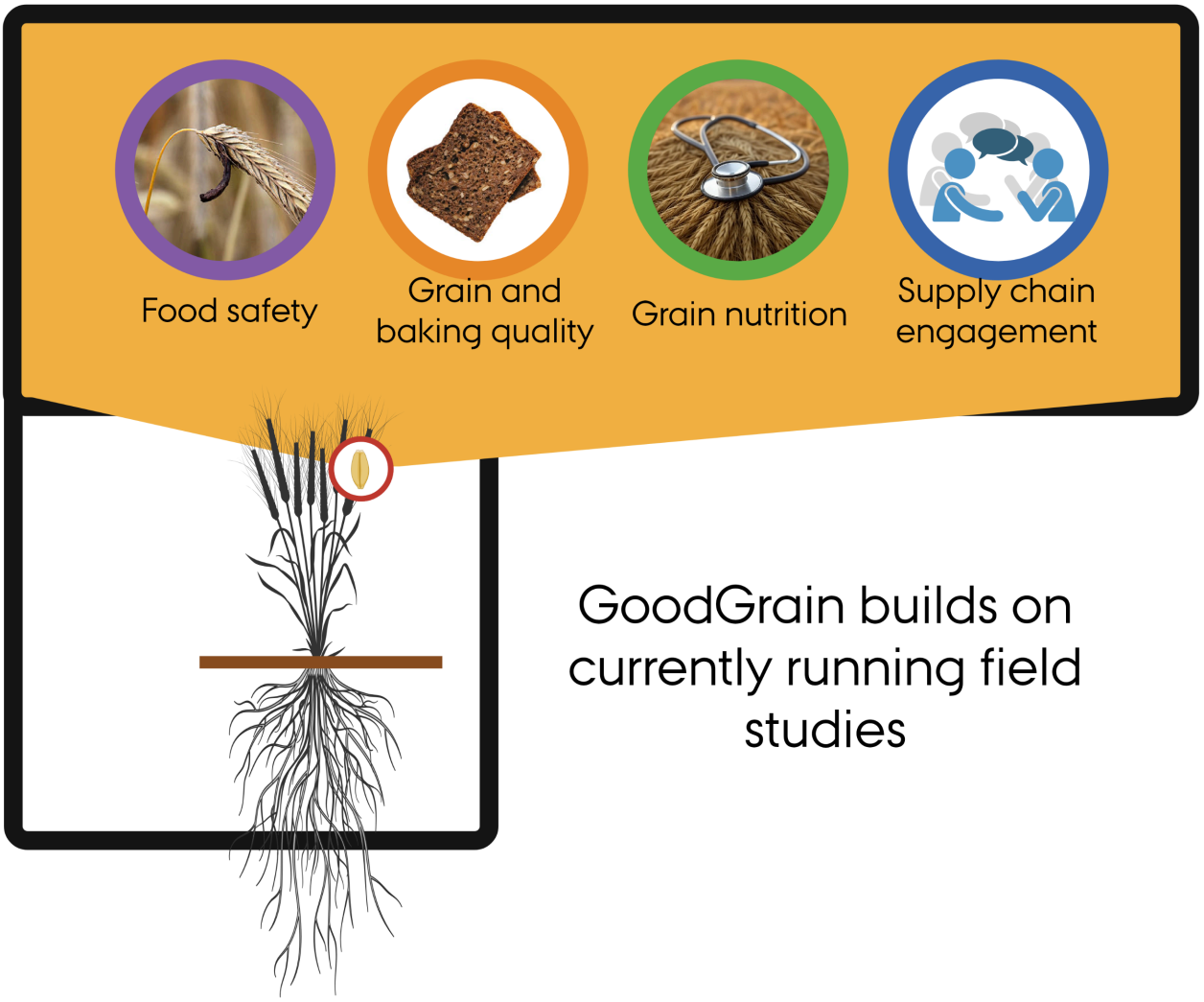
What happens next?

1. Read this Informational Sheet (OR watch the video).
2. After reading (or watching the video), if you decide you would like to participate in our research, *complete the intake form* using the link that was emailed to you.

An overview of the GoodGrain project

All of the crops commonly used to make bread are annuals (for example wheat, oats, and rye). This means they are sown, produce one grain harvest per year, then must be sown again. If a crop were perennial and lived for several years, it could produce two or three grain harvests from one sowing. This would save the farmer a lot of work (and tractor driving). Furthermore, by allowing the crop to remain in the field for many years without disturbing the soil, the perennial grain could potentially develop more roots, which would be better at taking up nutrients and water, would sequester more carbon and would feed the soil biology. There is more information presented about the idea and the history of perennial grains in the '***For the Curious***' section.

A few varieties of perennial grains have recently become available, including a perennial rye grain. This new variety is being tested in the field to evaluate how it grows in Denmark, as well as how it impacts the environment as compared to an annual rye crop. The **GoodGrain** project is meant to build on that knowledge by looking specifically at the new perennial rye's grain characteristics.



One of the perennial rye grain characteristics GoodGrain is studying is baking quality, including how it performs in home baking applications. The objectives of this home baking study are to:

1. Start to understand the **best uses** for this perennial rye for home bakers
2. Identify some potential perennial rye flour performance areas that require **further research**
3. Understand **how people feel** about baking with perennial rye

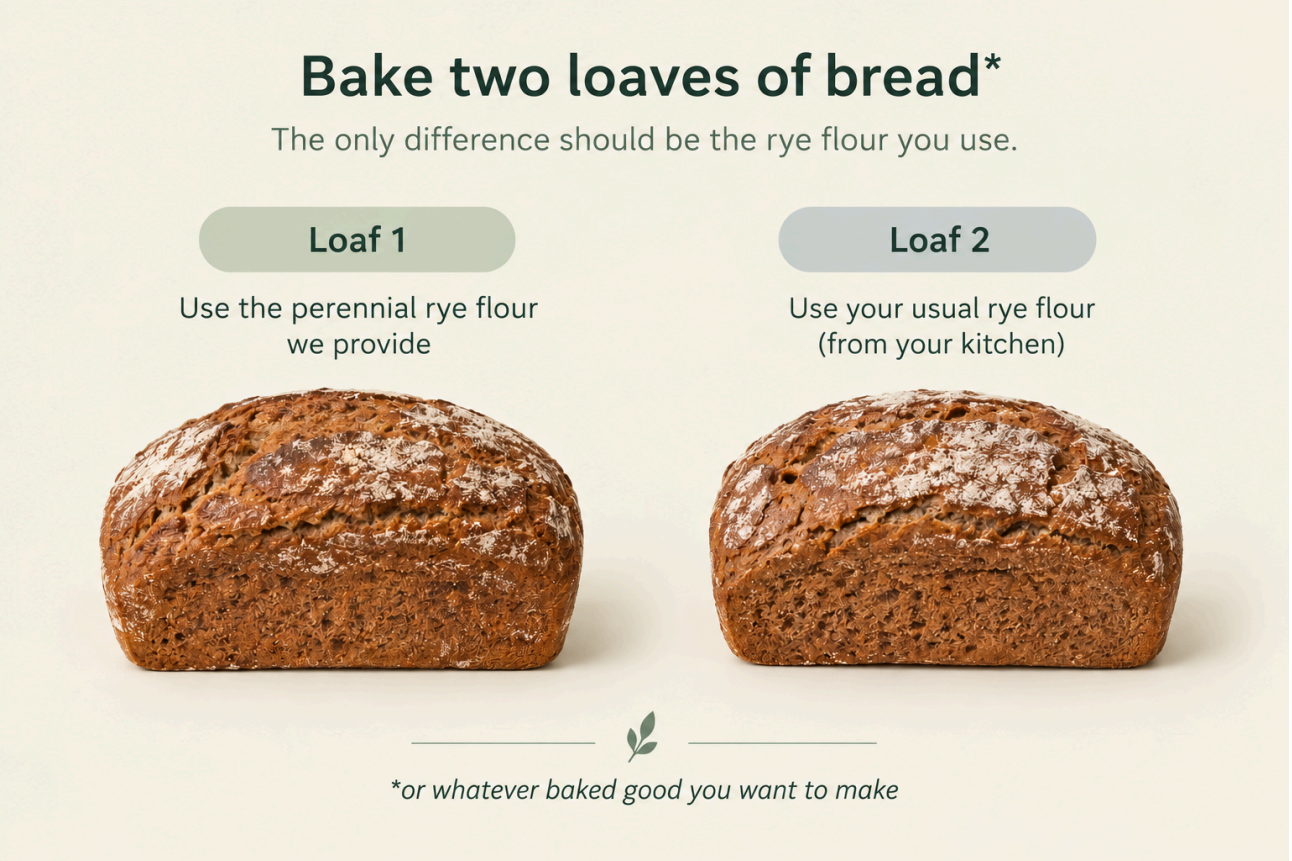
It is important to the **GoodGrain** project to engage with our communities in this effort. You can read a bit more about the GoodGrain project here.

What is expected of me as a participant in the GoodGrain home baking study?

Make two versions of a baked good (One or more days)

As a citizen scientist home baker, we will provide you with either 400 or 800 grams of flour (based on your use needs). Our hope is that you will do a **side-by-side baking experiment**, using the **perennial rye flour** we provide and the **rye flour you normally use**.

You are welcome to mix the rye flour with other grains or flours if that is what you normally do. If you use a rye sourdough starter, you can try to use this rye in it. If you normally make crackers, or pizza dough, or cookies, that is fine too! **We want to you to feel free to use the flour to make whatever brings you joy!** But we ask that you try to be as scientific as you can, and bake your creation with both the perennial rye, and your normal rye.



Our hope is that you will do a side-by-side bake, using the perennial rye we provide and the rye you normally use. Image made with the help of AI.

Note that you will need to purchase any additional ingredients you need to do your baking.

Take notes

You will be provided with a sheet to guide your note-taking during your baking, which you can use to fill out the post-baking survey.

Fill out post-baking survey (15-30 minutes)

In the post-baking survey, there are three general sections:

1. Simple **demographic** information about everyone who was involved in the baking (we know some people may bake on their own, while others may bake with their kids, parents, or partners)
2. Questions about any **differences** you noticed between the perennial rye product and the one made with the annual rye
3. Questions about your **experience**

We estimate the post-baking survey will require between 15-30 minutes to fill out. **Depending on what you chose to bake, this project could take one or more days to complete!**

We understand this is a big time commitment, and upon completion of the post-baking survey, you will be sent a small thank you gift 🎁.

Optional activities

In the summer of 2027, the **GoodGrain** project will also host an event where you can see the perennial rye variety growing in the field in a test plot. In the fall of 2027, there will be a public symposium in Copenhagen to share current knowledge about perennial rye with everyone across the value chain (researchers, breeders, farmers, millers, bakers, and bread eaters). The events will be free and open to the public - we welcome you to attend, but your attendance is completely *optional*.

Answers to some common questions

Is it safe to eat?

Yes, the grain used to make the flour you are receiving was tested at an independent laboratory to ensure it is safe for human consumption, as well as to measure the protein content. You can access the lab results (in Danish) here: GoodGrain Lab Results (PDF).

Is this perennial rye variety genetically modified using new breeding technologies?

No, this variety was developed using traditional breeding methods, involving crossing plants by hand, and making selections based on their physical characteristics. You can read more about it in this scientific publication from 1986.

Is this grain organic?

No, Aarhus University does not have any organically certified land due to the costs and additional administrative complexity that would entail. However, this grain was grown without the use of pesticides.

Can I buy this flour somewhere?

No, flour from this perennial rye variety is not currently available commercially. This project is laying the groundwork for a potential commercial release, and your help is appreciated!

Funding and project partners

This project is part of a **Novo Nordisk Fonden Plant2Food** research project (Grant #NNF22SA0081019) being conducted at the Department of Agroecology and Department of Food Science at **Aarhus University** in Denmark in collaboration Wageningen University

and Research in the Netherlands, and with Meyers, a prominent Danish food enterprise. As mentioned above, you can read a bit more about the GoodGrain project here.